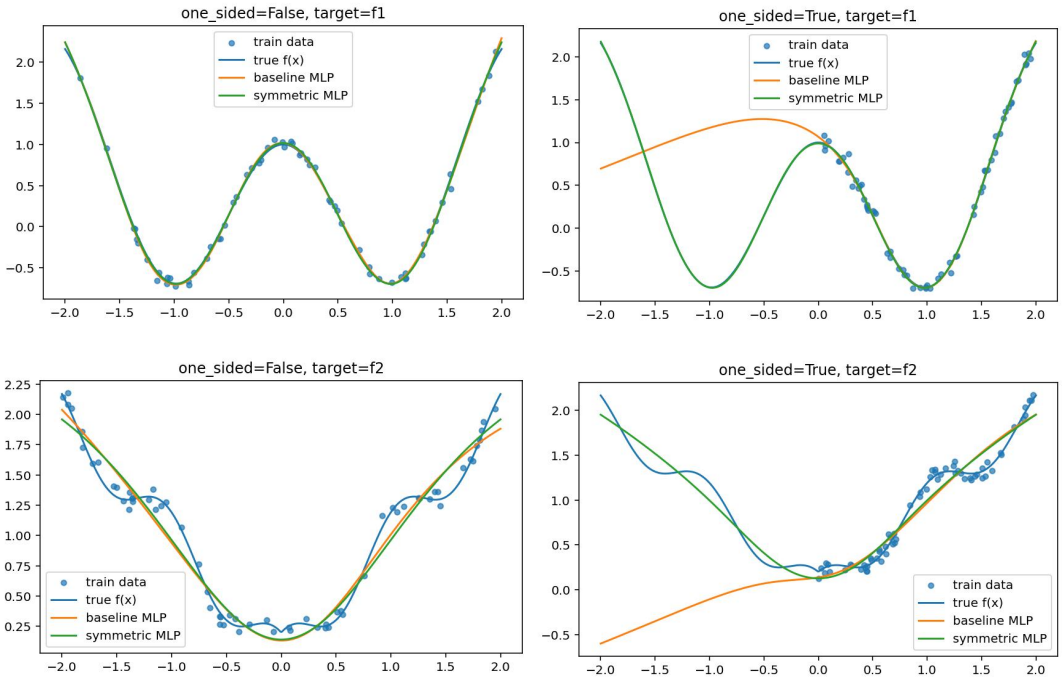
Problem 1:

Use MLP as the baseline g(x). Build up the symmetric model with h(x)=0.5*(g(x)+g(-x)). Compare the symmetric model and the baseline with even functions cos(3x)+0.3x^2 and 0.2cos(6x)+|x|. The results show that when the train data are balanced, including both positive and negative x values, two models give similar predictions. When train on only positive x values, the symmetric models predict much better than the baseline.



Problem 2:

Part 1:

The Gray-Scott reaction diffusion system is used in this problem, with the coupled PDEs:

$$\frac{\partial u}{\partial t} = D_u \nabla^2 u - uv^2 + F(1-u)$$

$$\frac{\partial v}{\partial t} = D_v \nabla^2 v + uv^2 - (F+k)v$$

The trained dataset is the subset of gray_scott_reaction_diffusion dataset from The Well, with 80 train data, 20 valid data, and 20 test data. The simulations are defined on a uniform Cartesian grid with periodic boundary conditions.

The enforced symmetry is translation equivariance. The Gray-Scott system has periodic boundary conditions, so the local evolution law does not depend on absolute positions. If the input field is shifted in space, the predicted field will shift in the same way.

Part 2:

Convolutional neural network is used in this problem. Use the zero-padding CNN as the baseline and circular-padding CNN as the constrained architecture. A simple architecture is:

$$h^{(1)} = A(W^{(1)} * x + b^{(1)})$$

$$h^{(2)} = A(W^{(2)} * h^{(1)} + b^{(2)})$$

$$y = W^{(3)} * h^{(2)} + b^{(3)}$$

Here, the input x is is the field configuration at time t, output y=f(x) is the predicted field at the next time step, * denotes 2D convolution, $W^{(i)}$ are learnable convolution kernels, $b^{(i)}$ are bias

terms, and A is a pointwise activation function ReLU. Circular padding is used in every layer so that the architecture matches the periodic boundary conditions of the dataset. The architecture is translation equivariant, meaning that for any spatial translation T:

$$f(Tx) = Tf(x)$$

Consider one convolutional layer:

$$L(x) = W*x + b$$

If the input is translated, then:

$$L(Tx) = W*(Tx) + b$$

With the definition of convolution:

$$W*(Tx) = T(W*x)$$

The bias is added equally at every spatial position, it does not break translation equivariance. Therefore:

$$L(Tx) = TL(x)$$

the activation function A is applied pointwise, so it also commutes with translation:

$$A(Tx) = TA(x)$$

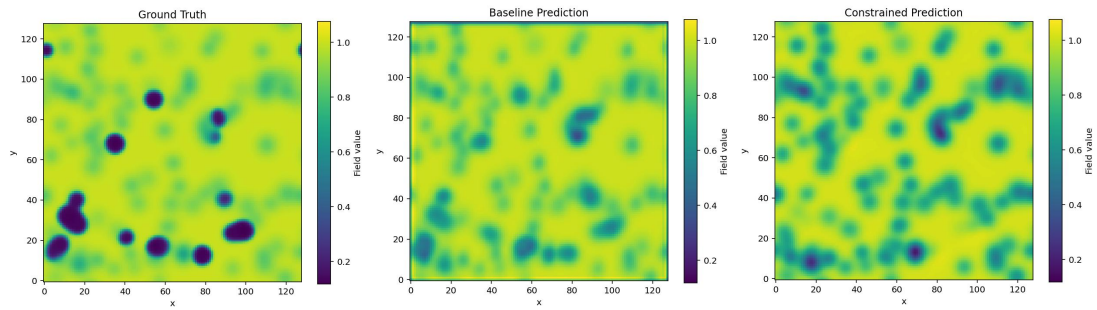
Each convolutional layer is translation equivariant and each activation function is also translation equivariant, the composition of these layers remains translation equivariant. Therefore, the full network satisfies:

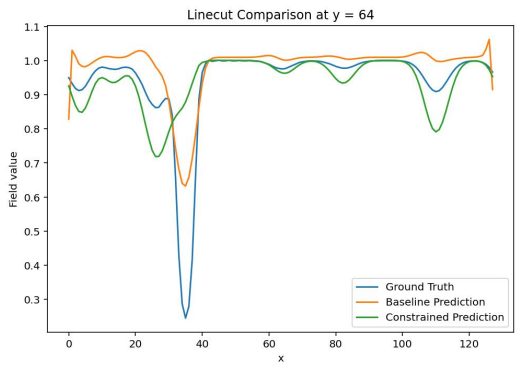
$$f(Tx) = Tf(x),$$

Which is the translation equivariant.

Part 3:

The results are shown as attached below:






```
Pre-Training Translation Equivariance Sanity Check:
Baseline relative error: 1.364467e-01
Constrained relative error: 2.229444e-07

Training baseline model (zero padding):
[Baseline] Epoch 001 | train loss=0.077070 | valid MSE=0.019515
[Baseline] Epoch 002 | train loss=0.004943 | valid MSE=0.004293
[Baseline] Epoch 003 | train loss=0.001630 | valid MSE=0.002853

Training constrained model (circular padding):
[Constrained] Epoch 001 | train loss=0.132683 | valid MSE=0.010916
[Constrained] Epoch 002 | train loss=0.002920 | valid MSE=0.004156
[Constrained] Epoch 003 | train loss=0.000996 | valid MSE=0.003981

Final Test Results:
Baseline    | test MSE=0.001342
Constrained | test MSE=0.001088

Post-Training Translation Equivariance Sanity Check:
Baseline relative error: 5.613641e-02
Constrained relative error: 9.489632e-08
```

Since the zero padding will induce boundary effects and breaks equivariance at boundaries, thus baseline model is not strictly translation equivariant compared to constrained model. This is validated in the sanity check. The constrained model gives smaller MSE compared to baseline, which outperform a better prediction. Yet the constrained model is more time-costing, due to more complex boundary conditions.